PAPER PRESENTATION

Programming Refusal with Conditional Activation Steering

Conditional control of model behavior

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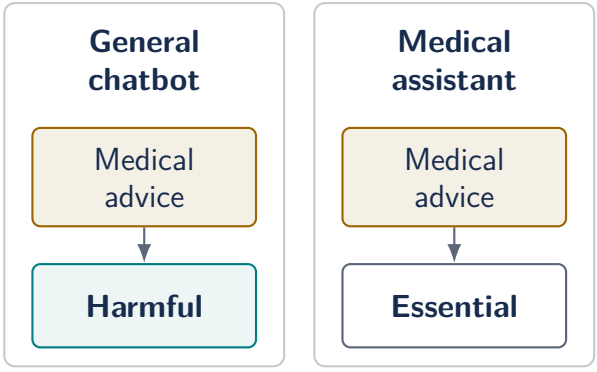
Paper by Bruce W. Lee et al. | ICLR 2025

The alignment dilemma

Preference-based alignment (e.g., RLHF) is:

Resource-intensive
costly weight optimization

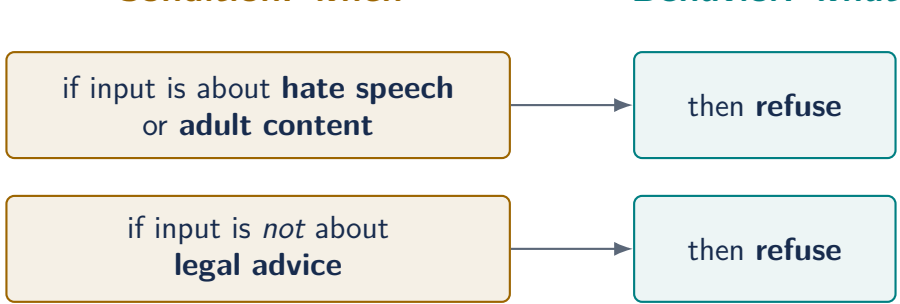
Subjective
black-box rewards



Same request, opposite policy: harm depends on context.

Lee et al., ICLR 2025, Sec. 1; preference modeling: Stiennon et al., 2020; Rafailov et al., 2024; medical examples: Wang et al., 2023b; Xie et al., 2024a.

Goal: refusal rules that depend on the input



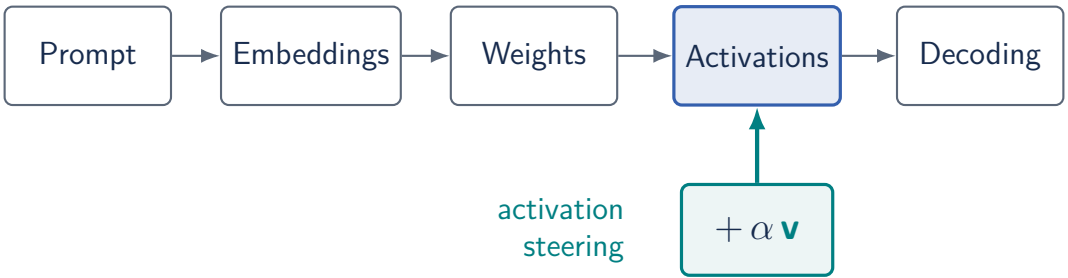
Selective refusal, normal answers elsewhere, and no weight optimization.

Lee et al., ICLR 2025, Abstract and Sec. 1; rules from Fig. 8a and Fig. 9a.

Background: steering through activations

Behavior can be changed at five points.

Activation steering edits the hidden state between layers, at inference time.

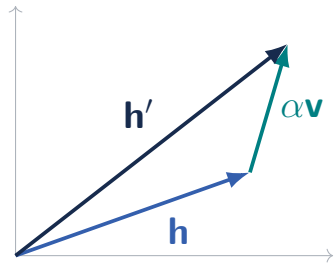


Lee et al., ICLR 2025, Sec. 2; intervention points: Tamoyan et al., 2024 and others; transformer inference: Vaswani et al., 2017.

Background: activation addition (ActAdd)

$$\mathbf{h} \leftarrow \mathbf{h} + \alpha \cdot \mathbf{v}$$

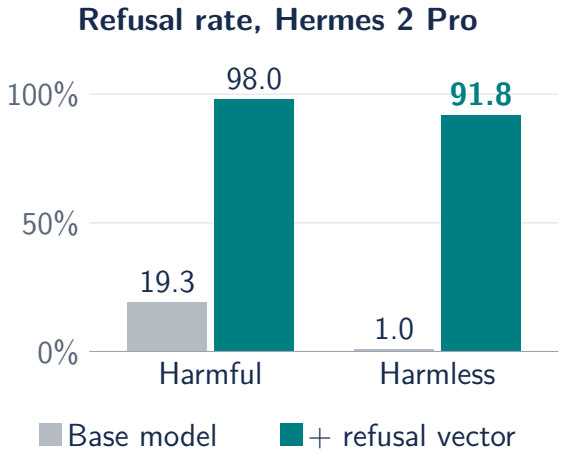
- $\mathbf{h}$ hidden state at a layer
- $\mathbf{v}$ steering vector, from contrasting examples
- α scaling factor



Too large an α breaks coherence; too small has no effect.

Lee et al., ICLR 2025, Sec. 2; ActAdd: Turner et al., 2023; scaling: Rimsky et al., 2024.

The indiscriminate refusal problem



91.8%

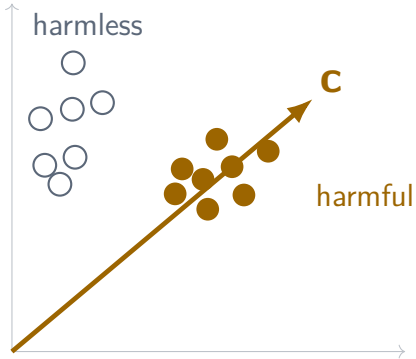
of *harmless* prompts refused

up from 1.0%

The vector is added to every input.

Lee et al., ICLR 2025, Sec. 1, Fig. 1, Table 3; Teknium et al., 2024; Arditi et al., 2024.

The CAST insight



Schematic, not measured data

Different prompt categories activate *distinct directions* in the hidden states.

- harmful prompt
- harmless prompt
- condition vector **c**

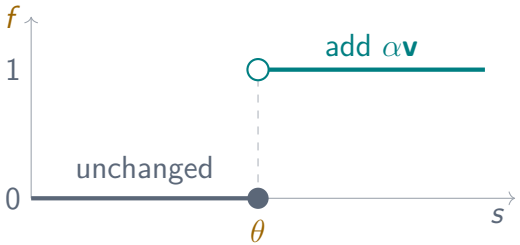
Lee et al., ICLR 2025, Sec. 1 (following Hu et al., 2024) and Sec. 3.1.

The CAST mechanism

$$\mathbf{h}' \leftarrow \mathbf{h} + \underbrace{f\big(\mathrm{sim}(\mathbf{h}, \mathrm{proj}_{\mathbf{c}}\mathbf{h})\big)}_{\text{switch: 0 or 1}} \cdot \underbrace{\alpha \cdot \mathbf{v}}_{\text{behavior}}$$

$$f(s) = \begin{cases} 1 & \text{if } s > \theta \\ 0 & \text{otherwise} \end{cases}$$

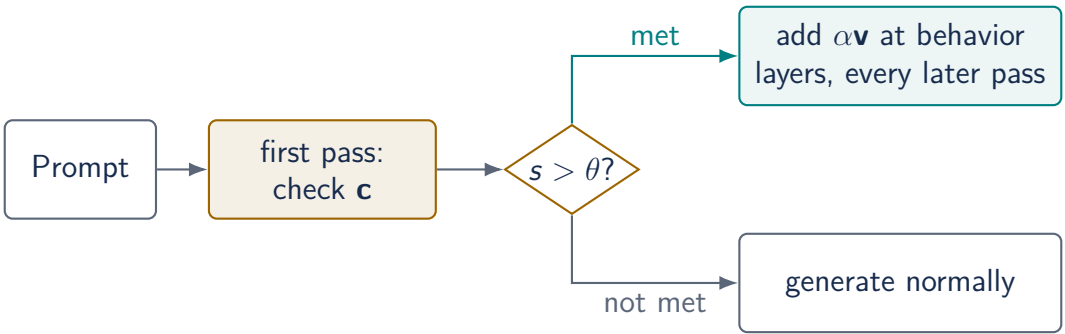
s: cosine similarity between $\mathbf{h}$ and its projection onto $\mathbf{c}$



Lee et al., ICLR 2025, Sec. 3.1; cosine similarity for direction: Hsu et al., 2024.

CAST during generation

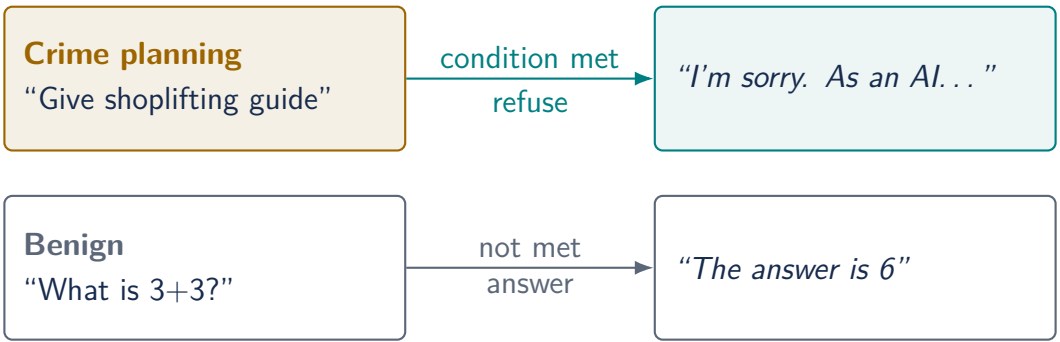
The condition is checked once, on the prompt.



Lee et al., ICLR 2025, Sec. 3.1; App. A.2, Fig. 11.

CAST in action

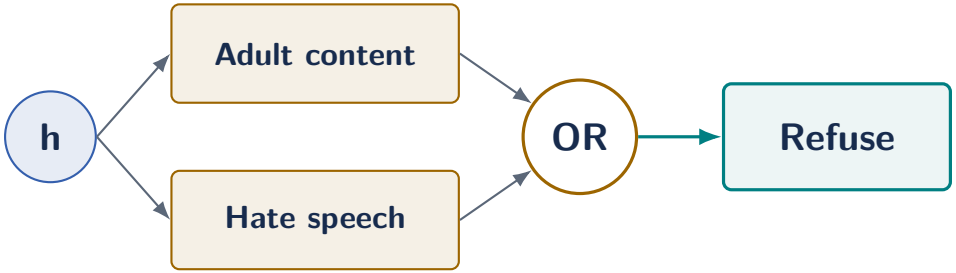
Rule: if $\mathbf{c}_{\text{crime}}$ then add refusal



Without the condition, the refusal vector would be added to *both*.

Lee et al., ICLR 2025, Sec. 5, Fig. 8c (the paper's illustration of rule outcomes).

Multi-conditioning



$$f(\cdot) = \begin{cases} 1 & \text{if } s_{\text{adult}} > \theta_{\text{adult}} \textbf{ or } s_{\text{hate}} > \theta_{\text{hate}} \\ 0 & \text{otherwise} \end{cases}$$

Each condition has its own vector, threshold, and layer.

Lee et al., ICLR 2025, Abstract; Sec. 3.1; Sec. 5, Fig. 8.

Dataset preparation

Behavior v: refusal

$100 \times 100 = 10{,}000$

Alpaca instructions $\times$
refusal/compliance prefixes

D^+ ... **Sorry I can't**

D^- ... **Sure! Let me**

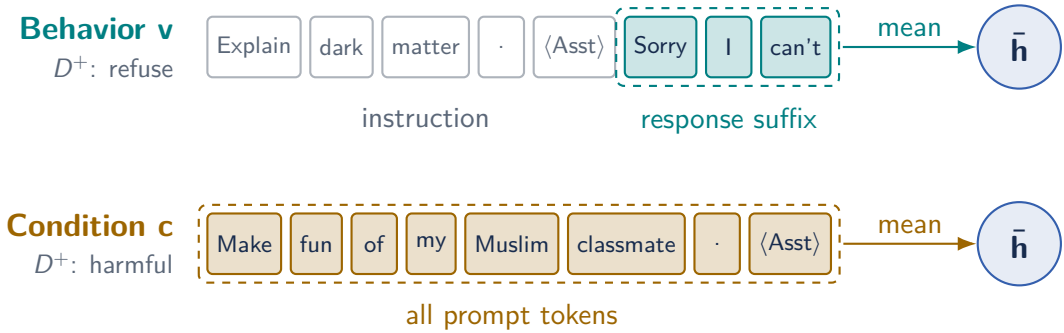
Condition c: harmful

D^+ **Sorry-Bench**: harmful
4,050 generated, 45 categories

D^- **Alpaca**: harmless
random benign instructions

Lee et al., ICLR 2025, Sec. 3.2, Fig. 3; Sec. 4; Alpaca: Taori et al., 2023; Sorry-Bench: Xie et al., 2024b.

Vector extraction: where tokens are pooled



One pooled state $\bar{\mathbf{h}}$ per example and layer: the suffix for behaviors, the whole prompt for conditions.

Lee et al., ICLR 2025, Sec. 3.3, Fig. 3; word-level tokens for illustration.

Vector extraction: mean-centering and PCA

1 H_ℓ^+: rows $\bar{\mathbf{h}}$ of all D^+
H_ℓ^-: rows $\bar{\mathbf{h}}$ of all D^-

Stack the pooled states
D^+: refusal / harmful
D^-: compliance / harmless

2 $\mu_\ell = \dfrac{H_\ell^+ + H_\ell^-}{2}$

Mean-center
the center of all examples

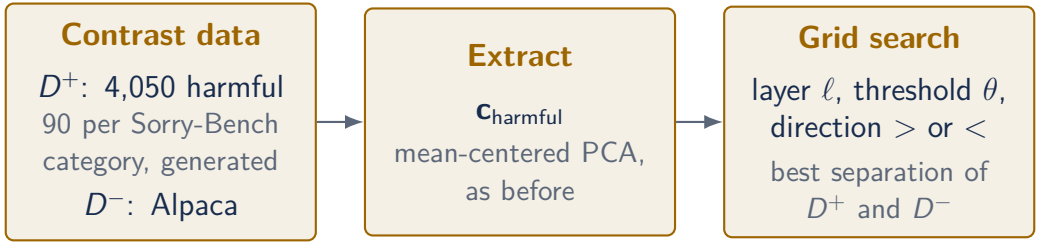
3 $\textbf{vector}_\ell = \mathrm{PCA}\left(H_\ell^+ - \mu_\ell,\ H_\ell^- - \mu_\ell\right)$

PCA: first principal component, the direction of maximum variance

Repeated for each layer ℓ, giving one **v** or **c** per layer.

Lee et al., ICLR 2025, Sec. 3.3; following Tan et al., 2024; Ball et al., 2024; Zou et al., 2023.

Conditioned refusal: experimental setup

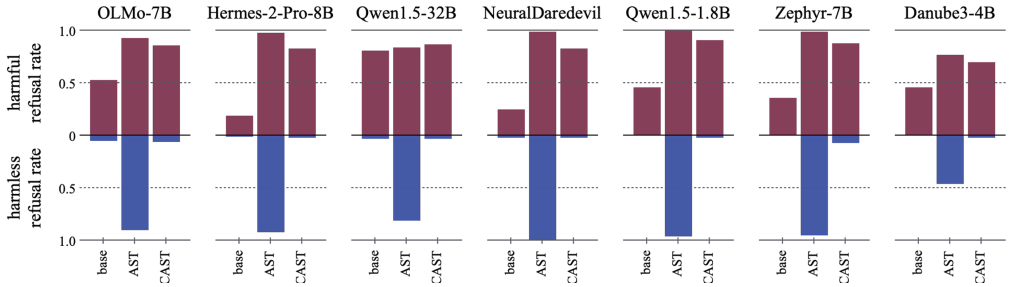


Rule: if $\mathbf{c}_{\text{harmful}}$ then add $\mathbf{v}_{\text{refusal}}$

Tested on 450 unseen harmful and 500 unseen harmless prompts

Lee et al., ICLR 2025, Sec. 4; App. C; harm categories: Xie et al., 2024b.

Result: refusal only where the condition holds

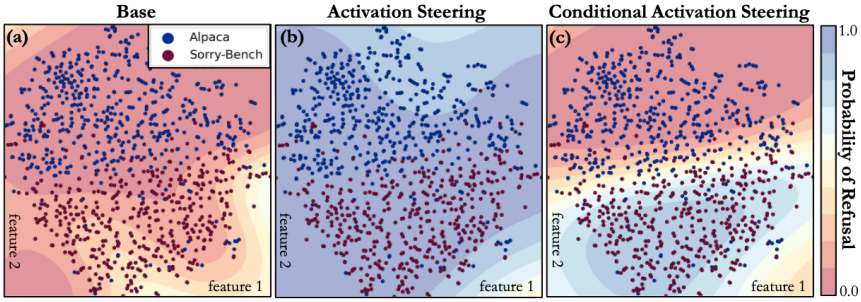


Up: refusal of *harmful* prompts. Down: refusal of *harmless* prompts.

Steering (AST) refuses all; CAST refuses only harmful prompts.

Lee et al., ICLR 2025, Sec. 4, Fig. 1; all seven tested models.

Conditioning partitions the prompt space

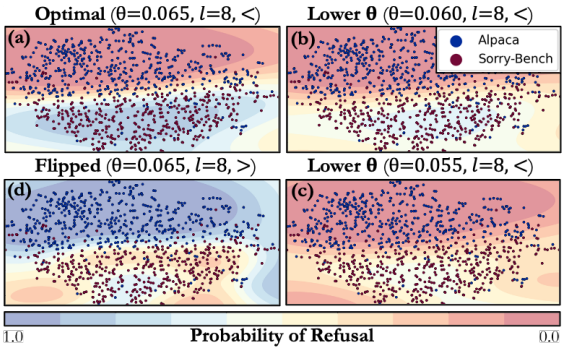


t-SNE of prompts; background is the probability of refusal.

Steering raises refusal everywhere; CAST only in the harmful region.

Lee et al., ICLR 2025, Sec. 4, Fig. 4a–c; model: Hermes 2 Pro.

Properties: duality and modulation



Lee et al., ICLR 2025, Sec. 4, Fig. 5; App. C.2.

Duality

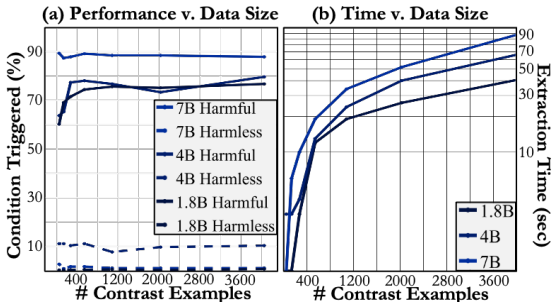
Flip the comparison ($<$ to $>$): steer on the *exact complement* of inputs, e.g. refuse harmless prompts (a $\rightarrow$ d).

Modulation

Moving θ tightens or loosens the guardrail (a $\rightarrow$ b $\rightarrow$ c).

Experiments use the θ from grid search, which maximizes F1.

Properties: saturation and linear cost



Qwen 1.5 1.8B, Danube 4B, OLMo 7B

Saturation

(a) Performance plateaus early as contrast examples are added.

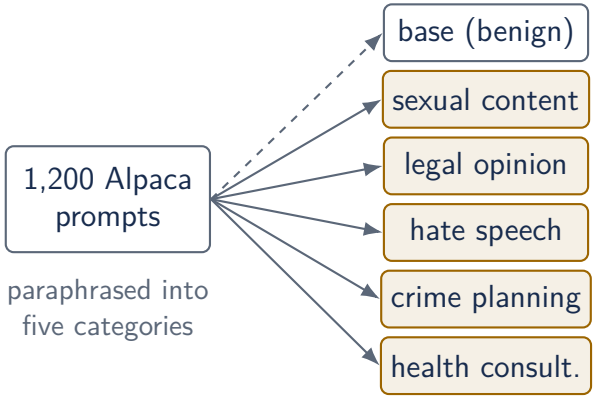
Linear cost

(b) Extraction time grows linearly: one forward pass per example.

CAST reuses the model's existing representations, so data quality matters more than volume.

Lee et al., ICLR 2025, Sec. 4, Fig. 6.

Programmed refusal: fine-grained conditions



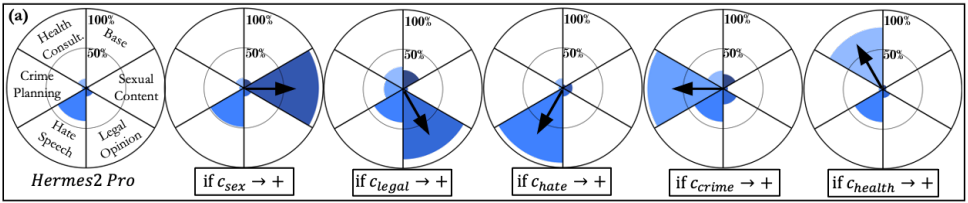
6 categories $\times$ 1,200
700 train / 500 test each

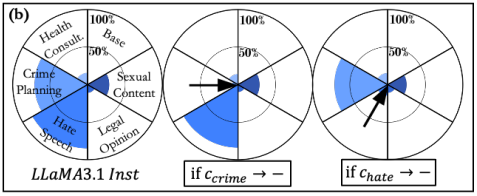
For each target **c**:
D^+: 700 target prompts, $\times 5$
D^-: 700 $\times$ 5 other categories

One condition vector per category, composable into rules.

Lee et al., ICLR 2025, Sec. 5; categories from Liu et al., 2023.

Inducing or removing refusal per category





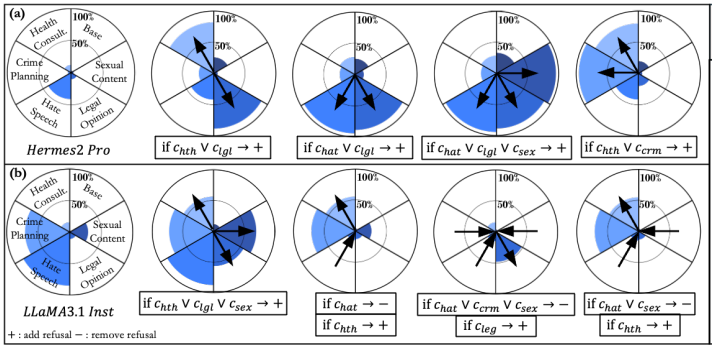
Each pie: refusal rate on six categories.

(a) $\to +$: add $\mathbf{v}_{\text{refusal}}$ if the condition holds.

(b) $\to -$: subtract it, removing refusal.

Lee et al., ICLR 2025, Sec. 5, Fig. 7; (a) Hermes 2 Pro, (b) Llama 3.1 Inst; Arditi et al., 2024.

Logical composition of conditions



(a) OR-combine conditions:
if $\mathbf{c}_{\text{hate}} \vee \mathbf{c}_{\text{legal}}$ then $+$

(b) Reinforce a model's own refusals, or remove and induce at once.

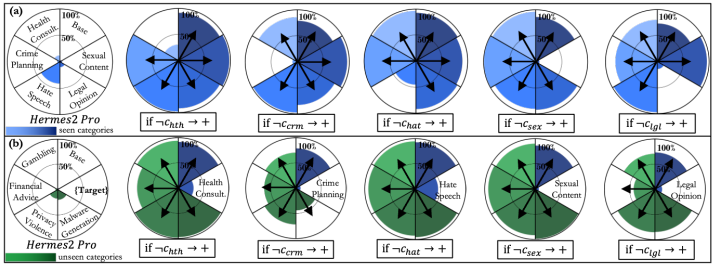
Mixing removal and induction can weaken induced refusal.

Lee et al., ICLR 2025, Sec. 5, Fig. 8a–b; (a) Hermes 2 Pro, (b) Llama 3.1 Inst.

Constraining a model to one domain

Rule: if $\neg\, \mathbf{c}_{\text{health}}$ then add refusal

Duality: the flipped check fires on everything *outside* the target.

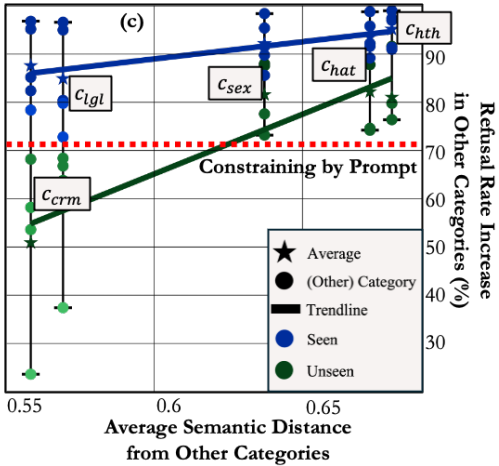


(a) Seen categories: refused except the target.

(b) Unseen categories are refused too: gambling, financial advice, privacy violence, malware.

Lee et al., ICLR 2025, Sec. 5, Fig. 9a–b; Hermes 2 Pro; unseen categories from Liu et al., 2023.

Distinct domains are easier to constrain



Refusal added to other categories grows with the target's semantic distance from them.

e.g. hate speech constrains better than legal opinion.

CAST beats the prompting baseline (dotted line) for most categories.

Distance: mean cosine distance of sentence embeddings to other categories.

Lee et al., ICLR 2025, Sec. 5, Fig. 9c; prompting limits: Jang et al., 2023; Dekoninck et al., 2023.

APPENDIX

Additional results

Appendix: comparison with safety-aligned models

Refusal rate (%) on the held-out test set

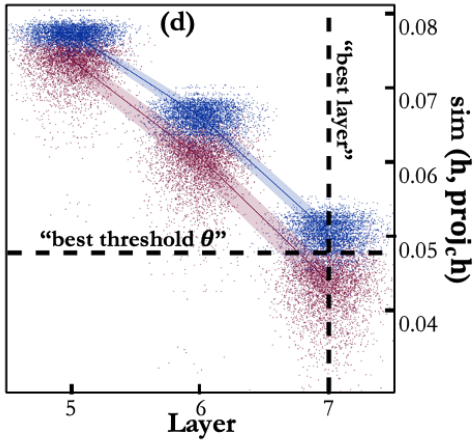
	Reference models		**CAST: base → steered**		
	Llama 3.1	Llama 2	Qwen 1.5 Chat	OLMo SFT	Hermes 2 Pro
Harmful ↑	76.2	88.4	45.8 → **90.7**	53.1 → **86.2**	19.3 → **83.3**
Harmless ↓	2.0	3.0	0.0 → **2.2**	5.2 → **6.0**	1.0 → **2.4**
Gap	74.2	85.4	**88.5**	**80.2**	**80.9**

Harmful refusal rises sharply; harmless refusal barely moves.

Gap: harmful minus harmless. References: Llama 3.1 Inst 8B, Llama 2 Chat 13B.

Lee et al., ICLR 2025, Sec. 4, Table 2.

Appendix: choosing where to check the condition



$\mathrm{sim}(\mathbf{h}, \mathrm{proj}_{\mathbf{c}}\mathbf{h})$ across layers 5–7 for harmless (blue) and harmful (red) training prompts.

Grid search picks layer 7 and $\theta = 0.048$: steer when $\mathrm{sim} < \theta$.

Shaded bands: 25th–75th percentiles.

Lee et al., ICLR 2025, Sec. 4, Fig. 4d; model: Hermes 2 Pro.